Advance Mobile Application Development

File management in flutter

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Read and write files

In some cases, you need to read and write files to disk. For example, you might need to persist data across app launches, or download data from the internet and save it for later offline use.

To save files to disk on mobile or desktop apps, combine the path_provider plugin with the dart:io library.

This recipe uses the following steps:

- Find the correct local path.
- Create a reference to the file location.
- Write data to the file.
- Read data from the file.

Working with Excel Files in Flutter

- The first thing we need to do is to add excel package to our pubspecp.yaml file.

```
excel:
path:
permission_handler:
file_picker:
```

```
syncfusion_flutter_xlsio:
```

- And import their packages.

```
import 'package:flutter_excel/excel.dart';
import 'package:permission_handler/permission_handler.dart';
import 'package:path/path.dart';
```

Create New XLSX File

```
var excel = Excel.createExcel();
```

Which will automatically creates 1 empty sheet: Sheet1

Inserting data into an excel file

Now that we have an excel file, we can move on to inserting files into the file. there are two ways of inserting data into an excel file:

1. **Getting reference to a single cell in the a sheet**
2. **Inserting values into a row with List<String>**

Getting reference to a single cell in the a sheet

```
Sheet sheetObject = excel['فاتورة 01'];
var cell = sheetObject.cell(CellIndex.indexByString("A1"));
cell.value = 8; // dynamic values support provided;
```

1. **Sheet**: A sheet is a single page inside of an excel file, if you are familiar with excel you will understand that a single excel file can contain multiple sheets
   1. **excel[‘**فاتورة 01**’]:** Each sheet in an excel file has a name, so to get reference to the sheet you need it’s name
2. In the second line of code is used to get reference to cell “A1”, which as we know is the first cell in the sheet
3. This line sets the value of cell “A1” to 8

Inserting values into a row with List<String>

```
Sheet sheetObject = excel['فـاتـورة 01'];
List<String> data = ["Mr","’mazin", "Alkathiri"];
sheetObject.appendRow(data);
```

- **sheetObject.appendRow(date):** This Line of code inserts the List values into the sheet like this:

createexcelfile()

```
void createexcelfile() async{
  Sheet sheetObject = excel['فـاتـورة 01'];
  var cell = sheetObject.cell(CellIndex.indexByString("A1"));
  cell.value = 8; // dynamic values support provided;
  List<String> data = ["Mr","'mazin", "Alkathiri"];
  sheetObject.appendRow(data);
  saveExcel();
}
```

saveExcel()

```
saveExcel() async {
  //request for storage permission
  var res = await Permission.storage.request();

  var fileBytes = excel.save();
  File(join("/storage/emulated/0/Download/billNo03.xlsx"))
    ..createSync(recursive: true)
    ..writeAsBytesSync(fileBytes!);
}
```

reading XLSX File

```
var bytes = file!.readAsBytesSync();
var excel = Excel.decodeBytes(bytes);
```

- readAsBytesSync abstract methodSynchronously reads the entire file contents as a list of bytes.

reading XLSX File

```
for (var table in excel.tables.keys) {

  for (var row in excel.tables[table]!.rows) {
            for (var cell in row) {
                        print(cell!.value);
                                    }
                                                }
                                          }
```

void readexcelfile(File? file) async

```
void readexcelfile(File? file) async {
  var bytes = file!.readAsBytesSync();
  var excel = Excel.decodeBytes(bytes);
for (var table in excel.tables.keys) {
int counter=0;
  for (var row in excel.tables[table]!.rows) {
    if(counter==1){
      int counter2=0;
      for (var cell in row) {
        if(counter2==4) {
          if (cell != null)
            print(cell!.value);
        }
        counter2++;
      }
    }
```

```
if(counter==2){
      int counter2=0;
      for (var cell in row)
{
        if(counter2==4) {
          if (cell != null)

print(cell!.value);
        }
        counter2++;
      }
    }
```

```
if(counter>6){
      int counter2=0;
      for (var cell in row) {
        if(counter2==1) {
          if (cell != null)
            print(cell!.value);
        }
        if(counter2==2) {
          if (cell != null)
            print(cell!.value);
        }
        counter2++;
      }
    }
    counter++;
  }
}
```

FilePicker

```
ElevatedButton(
  onPressed: () async {
    try {
      result = await FilePicker.platform.pickFiles();
      if (result != null) {
        if (!kIsWeb) {
          file = File(result!.files.single.path!);
          excelobj.readexcelfile(file);
        }
      } else {
        print("user cancled the picker");
      }

    } catch (_) {}
    super.setState(()  {
    });
  },
  child: const Text('Pick an Excel File'),
),
```